

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Applied Physics

Course Code: PHY-111

Examination: 2nd Periodical Examinations

Scheme: NEP

Branch: CSE/ECE/CE

Semester: 1st

Question Paper for 2nd Periodical Examination (Nov.- 2025) (Session Aug.-Dec. 2025)

Time: One and a half hours

Max Marks: 30

Note: Attempt all questions.

CO	Course Outcomes
CO1	Recognize the importance of Physics in describing different physical phenomenon
CO2	Comprehend the Special theory of relativity by examining inertial & non-inertial frames, Michelson-Morley experiment, Galilean & Lorentz transformations.
CO3	Demonstrate the concept of harmonic oscillators by studying simple harmonic, damped harmonic and forced harmonic oscillations.
CO4	Illustrate the concept of LASER, Optical Fibres and X-rays in various fields.
CO5	Appraise the concept of electromagnetic waves and Superconductivity in the field of Engineering
CO6	Recognize the inadequacy of classical mechanics for certain physical problems and thus find the solution of these problems using Quantum Mechanics

Que No.	Questions	Marks	Target CO	Target KL
1.(a)	Write momentum and energy operators.	1	CO6	L1
1.(b)	What are High Temperature Superconductors (HTSC)?	1	CO5	L1
1.(c)	Why wave nature of matter is not apparent to our daily observation?	1	CO6	L2
1.(d)	Discuss the need of quantum mechanics.	1	CO6	L2
1.(e)	What is meant by normalization of wave functions?	1	CO6	L1
2.	State and explain Heisenberg Uncertainty Principle. Using this principle show that electrons cannot exist in the nucleus.	5	CO6	L2
3.	Explain the de Broglie hypothesis of matter waves. Derive an expression for wavelength of an electron beam accelerated through a potential V.	5	CO6	L2
4.	State and explain Meissner effect. Discuss its contradiction with Maxwell's equation.	5	CO5	L3
5.	Explain isotope effect. How this effect gave the clue for the theory of superconductivity? Discuss BCS theory of superconductivity. Or Define phase velocity and group velocity. Show that the phase velocity of the de-Broglie wave is greater than velocity of light, but the group velocity is equal to the velocity of the particle with which the waves are associated.	10	CO5 / CO6	L2

Course Advisor: Dr. Neelam Guleria

Designation: AP (AS&H) Physics

CO: Course Outcome

KL: Knowledge Level: L1 – Remember L2 – Understand L3 – Apply L4 Analyze L5 – Evaluate L6 – Create